

Literature Benchmark Brief: Daenens_2025_WES

Title: Spatio-temporal graph neural networks for normal behaviour modelling and power loss estimation in offshore wind farms

Authors: S. Daenens, J. Helsen (2025)

Venue: Wind Energy Science | **DOI:** 10.5194/wes-10-385-2025

Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

Dynamic wake overlap adjacency weighting, Spatio-Temporal Graph Neural Network (STGNN)

Empirical Scale & Experimental Dataset

Belgian North Sea offshore wind farm, 30-second SCADA resolution

Relevance to IEEE TSTE & Sustainable Energy Systems

Proves superiority of GNN over shallow baselines in multi-turbine wake propagation

Identified Gaps & Operational Vulnerabilities

Evaluates purely intact SCADA without communication latency or packet dropouts

Role in Manuscript Defense & Reviewer #2 Proof

Defends why GNN outperforms GBDT at $h=6$: captures physical spatial wake diffusion